

About VALIDOS® — Validation Governance Architecture

Governing Validation Across the Complete Operational Lifecycle

Systems continuously produce claims.

AI systems produce outputs.

Autonomous systems make decisions.

Organizations rely on evidence.

Scientific environments test hypotheses.

Operational systems change.

Evidence changes.

Sources change.

Methods change.

Reliance changes.

Validation must remain governable.

VALIDOS® — Validation Governance Architecture was created to provide a stable validation governance architecture that remains applicable regardless of how technologies, AI systems, autonomous systems, scientific methods, organizations, regulatory frameworks, and operational environments evolve.

VALIDOS® does not define what is true, prescribe domain-specific validation criteria, replace scientific methods, determine regulatory compliance, certify products, prescribe testing technologies, or predetermine whether a Validation Object must receive a positive or negative outcome.

VALIDOS® governs validation.

Why VALIDOS® Exists

Most validation initiatives focus on testing, verification, evidence review, certification, quality assurance, model evaluation, scientific assessment, compliance, benchmarking, or technical performance.

These perspectives are valuable, but none of them alone answers one fundamental governance question:

How should validation itself be structured, governed, evidenced, executed, determined, relied upon, monitored, and repeated when reality changes?

VALIDOS® addresses this question by defining the governable validation spaces that exist throughout the complete validation lifecycle.

Rather than standardizing one validation method, **VALIDOS® standardizes the governance conditions under which Validation Objects can be identified, scoped, evaluated against governed criteria, examined through appropriate methods, supported by fit evidence and reliable sources, operationally validated, formally determined, assigned current Validation States, relied upon within defined boundaries, and revalidated when the basis supporting validation materially changes.**

Validation Before Reliance

The objective of VALIDOS® is not to prescribe how every validation must technically be performed.

The objective is to define how validation must remain governable.

Every validation requires a Validation Object.

Every Validation Object requires a defined purpose and scope.

Every validation scope requires applicable criteria.

Every criterion requires an appropriate validation method.

Every validation method requires evidence capable of supporting it.

Every material evidence basis depends upon sources whose reliability must be understood.

Every validation method must actually be executed.

Every execution must support a bounded Validation Determination.

Every Validation Determination must be translated into a current Validation State.

Every Validation State used operationally requires governed reliance.

Every material change affecting that validation basis may require revalidation.

VALIDOS® therefore treats validation as an independent governance domain.

The Validation Governance Architecture

VALIDOS® consists of ten independent validation governance spaces.

Each space answers one architectural question.

Each space governs one validation condition.

Each space has a defined beginning, operational purpose, and governance boundary.

Each space begins where the preceding validation governance requirement has been sufficiently established.

Together, these spaces form the complete validation governance architecture.

The architecture progresses through:

**Validation Object → Validation Purpose & Scope → Validation Criteria
→ Validation Method → Validation Evidence Fitness → Validation
Source Reliability → Validation Execution → Validation Determination
→ Validation State → Validation Reliance & Revalidation**

This progression represents the complete governance journey of validation—from identifying exactly what is being validated to governing whether, where, why, how long, and under which conditions the resulting validation may legitimately support operational reliance.

Validation Spaces Rather Than Technical Components

VALIDOS® does not organize validation governance around software products, testing platforms, laboratories, AI models, certification bodies, departments, vendors, databases, or implementation technologies.

Instead, it organizes governance around distinct validation spaces.

Each validation space represents an independent governance responsibility that exists regardless of any specific technology, platform, vendor, sector, validation technique, implementation method, or operational environment.

Because validation spaces are technology-neutral and domain-independent, **VALIDOS® remains applicable as AI capabilities, validation technologies, scientific methods, autonomous systems, regulatory environments, operational models, and governance requirements continue to evolve.**

A Methodology Rather Than an Implementation

VALIDOS® is a validation governance methodology.

It is not a technical testing implementation.

The architecture defines which validation conditions must exist, how they relate to one another, where validation governance responsibility begins and ends, what must remain traceable, and what

must be sufficiently demonstrated before stronger validation claims or operational reliance can be justified.

Organizations remain free to determine how validation is implemented within their own technical, scientific, operational, organizational, legal, and regulatory environments.

This separation allows **VALIDOS®** to remain stable while validation technologies, AI systems, scientific methods, operational processes, and implementation techniques continue to evolve.

From Validation Object to Governed Reliance

Validation cannot be governed as a single undivided act.

Before validation begins, the Validation Object must be identified.

Before criteria can be applied, Validation Purpose and Scope must be established.

Before evaluation occurs, governed Validation Criteria must exist.

Before evidence can be interpreted, an appropriate Validation Method must be established.

Before evidence supports a conclusion, its fitness must be assessed.

Before material evidence is relied upon, the reliability of its sources must be understood.

Before a determination exists, validation must actually be executed.

Before a validation result becomes operational, a formal Validation Determination must be established.

Before that determination can support reliance, its current Validation State must be known.

Before reliance continues, its scope, purpose, conditions, restrictions, duration, dependencies, and revalidation triggers must remain governed.

VALIDOS® transforms these dependencies into a structured and navigable validation governance lifecycle.

Validation Is Not a Permanent Event

A central principle of VALIDOS® is that validation is not automatically permanent.

A Validation Object may change.

Its configuration may change.

Its environment may change.

Its evidence may change.

Its sources may become unreliable.

Its intended use may expand.

Its autonomy may increase.

Its consequence profile may change.

New contradictory evidence may emerge.

A Validation State that was legitimate yesterday may therefore become insufficient tomorrow.

VALIDOS® addresses this through a recursive validation lifecycle:

Validation → Validation State → Reliance → Monitoring → Material Change → Revalidation Trigger → Validation Delta → Lifecycle Re-Entry → Revalidation → New Validation Determination → New Validation State → Renewed / Restricted / Suspended / Withdrawn Reliance

Validation therefore remains connected to the reality upon which it depends.

Validation State Is Not Universal Reliance

VALIDOS® separates two concepts that are frequently collapsed:

Validation State

and:

Reliance Authorization

An object may be Validated within one scope while remaining unsuitable for another purpose.

A model validated for advisory use may not automatically be validated for autonomous execution.

A system validated for one population may not automatically be validated for another.

A component validated in one operational environment may not automatically remain validated after material environmental change.

Therefore:

Validation does not create unlimited permission to rely.

Reliance must remain bounded by the validation basis actually established.

Negative Validation Is Still Valid Validation

VALIDOS® does not require validation to produce a positive result.

A governed validation process may legitimately produce:

Positive

Conditional

Partial

Negative

Indeterminate

or:

Conflicted

determinations.

A validation methodology that can only practically produce a positive result cannot provide meaningful validation governance.

VALIDOS® therefore preserves negative evidence, contradictory evidence, uncertainty, incomplete evidence, methodological limitations, execution deviations, and unresolved conditions as part of the validation record.

A negative outcome may represent a successful validation process.

The purpose of validation governance is not to produce approval.

The purpose is to produce a defensible determination.

Validation Depth

Not every Validation Object requires the same validation intensity.

A low-consequence informational output and a high-consequence autonomous system should not necessarily require identical validation depth.

VALIDOS® therefore allows Validation Depth to reflect factors such as:

- consequence,
- uncertainty,
- autonomy,
- complexity,
- reversibility,
- evidence quality,
- operational exposure,
- dependency,

- and intended reliance.

Greater consequence may require deeper evidence, stronger methods, stronger independence, broader execution, or additional assurance.

Validation Depth allows VALIDOS® to scale without reducing validation to either excessive uniformity or arbitrary discretion.

The Recursive Validation Lifecycle

Traditional validation models may treat validation as:

Test → Pass → Deploy

VALIDOS® treats validation as a lifecycle.

The governed lifecycle is:

Identify → Scope → Define Criteria → Select Method → Qualify Evidence → Qualify Sources → Execute → Determine → Assign State → Govern Reliance → Monitor Change → Revalidate

When material change occurs, revalidation begins at the earliest materially affected validation layer.

The complete architecture therefore becomes recursive rather than merely sequential.

This is particularly important for AI systems, autonomous systems, dynamic software, continuously changing datasets, adaptive systems, and operational environments where the basis supporting validation may evolve after deployment.

Four-Layer Operationalization

The ten Parent Standards define the complete VALIDOS® reference architecture.

For operational implementation, the architecture may be compressed through the:

VALIDOS® — Four-Layer Validation Protocol™ (4LVP™)

The protocol organizes validation into four operational layers:

Layer 1 — Object & Scope Validation

Layer 2 — Evidence & Source Validation

Layer 3 — Execution & Determination Validation

Layer 4 — State & Reliance Validation

Each layer produces a governed output that becomes an input to the next layer.

The protocol does not replace VALIDOS®.

VALIDOS® is the reference architecture.

4LVP™ is the operational validation pathway.

Its central anti-bypass principle is:

No validation claim may exceed the weakest unresolved validation layer on which that claim materially depends.

This makes it possible to use individual validation layers modularly while protecting the methodological meaning of Full Protocol Conformity.

Validation Traceability

A trustworthy validation claim should be capable of being traced backward.

From:

Reliance

to:

Validation State

to:

Validation Determination

to:

Validation Execution to:

Validation Method

to:

Evidence

to:

Sources

to:

Validation Criteria

to:

Validation Purpose & Scope

and ultimately to:

the exact Validation Object that was actually validated.

This backward traceability forms one of the structural foundations of VALIDOS®.

Who May Benefit

VALIDOS® may support:

- organizations governing complex operational systems,
 - public institutions,
 - regulators,
 - validation and assurance teams,
 - enterprise architects,
 - AI system developers,
 - autonomous system developers,
 - scientific and research organizations,
 - testing and evaluation teams,
 - data governance teams,
 - model governance teams,
 - healthcare organizations,
 - financial institutions,
 - industrial operators,
 - critical infrastructure providers,
 - certification and conformity environments,
 - decision-system operators,
 - and multi-system or cross-organizational ecosystems.
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Relationship to Other OOF® Architectures

VALIDOS® governs validation.

It operates alongside complementary OOF® architectures.

GOA™ governs governance structure, authority, delegation, scope, and continuity.

ORA™ governs operational reality, operational validity, execution, and coordination.

OBIDENTITY® governs origin, identity, ownership, authority, provenance, transfer, and continuity.

AGA™ governs accountability relationships, responsibility allocation, delegation, and ownership structures.

INTEGROS® governs integrity throughout operational lifecycles.

AIG® governs AI behavior.

CLIA® governs cognition, reasoning, interpretation, and cognitive governance.

MGIA™ governs memory, memory continuity, and organizational memory.

ASGA™ governs autonomous systems, autonomous agents, and autonomous operational environments.

SIMULOS® governs simulation integrity and the conditions under which simulation-derived outputs may support operational reliance.

ART™ governs continuous auditability, traceability, and evidence throughout operational lifecycles.

Together, these architectures create a structured methodology for governing increasingly intelligent, autonomous, evidence-dependent, interoperable, and operational environments.

The Long-Term Vision

As AI systems, autonomous systems, scientific infrastructures, digital systems, regulatory ecosystems, and organizations become increasingly interconnected, validation governance will become as essential as identity, integrity, accountability, security, operational reality, and interoperability.

The long-term objective of **VALIDOS®** is to provide a common validation governance language through which organizations can identify what is being validated, define what a validation actually covers, determine what evidence and methods support it, understand the current Validation State, establish legitimate reliance boundaries, and detect when changed reality requires revalidation.

VALIDOS® is intended to make validation governable:

before validation begins,

during validation execution,

when conclusions are produced,

before operational reliance,

throughout continued reliance,

and:

whenever reality materially changes.

Canonical Principle

Systems, evidence, methods, technologies, operational environments, and reliance contexts will continue to evolve. Validation governance must remain stable. VALIDOS® therefore governs the validation conditions that must remain demonstrable regardless of future technologies, implementations, methods, and operational environments.

Foundational Principle

Validation becomes trustworthy when every material claim can be traced backward through reliance, state, determination, execution, method, evidence, sources, scope, and the exact object that was actually validated.

Canonical Closing Statement

VALIDOS® — Validation Governance Architecture provides the governance foundation through which Validation Objects can be formally identified, purposefully scoped, evaluated against governed criteria, examined through appropriate methods, supported by fit evidence and reliable sources, operationally validated, formally determined, assigned current Validation States, relied upon within explicit boundaries, monitored through changing operational reality, and revalidated whenever the basis supporting validation materially changes. By defining the governable spaces of validation and connecting them through a recursive lifecycle, **VALIDOS®** establishes a technology-neutral methodology for ensuring that validation remains bounded, traceable, methodologically defensible, operationally relevant, reliance-aware, and continuously governable throughout the complete lifecycle.

Related Documents

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® — Four-Layer Validation Protocol™](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

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Canonical Source

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